

$$FF_{16} = 15 \times 16^1 + 15 \times 16^0$$

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$$CO_{16} = 12 \times 16^1 + 0 \times 16^0$$

$$A2B_{16} = 10 \times 16^2 + 2 \times 16^1 + 11 \times 16^0$$

$$9F2C_{16} = 9 \times 16^3 + 15 \times 16^2 + 2 \times 16^1 + 12 \times 16^0$$

$$FFAA_{16} = 15 \times 16^3 + 15 \times 16^2 + 10 \times 16^1 + 10 \times 16^0$$

$$27_8 = 2 \times 8^1 + 7 \times 8^0$$

$$147_8 = 1 \times 8^2 + 4 \times 8^1 + 7 \times 8^0$$

$$6755_8 = 6 \times 8^3 + 7 \times 8^2 + 5 \times 8^1 + 5 \times 8^0$$

$$7777_8 = 7 \times 8^3 + 7 \times 8^2 + 7 \times 8^1 + 7 \times 8^0$$

$$3456_8 = 3 \times 8^3 + 4 \times 8^2 + 5 \times 8^1 + 6 \times 8^0$$

$$110_2 = 1 \times 2^2 + 1 \times 2^1 + 0 \times 2^0$$

$$1011_2 = 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$$

$$10101010_2 = 1 \times 2^7 + 1 \times 2^5 + 1 \times 2^3 + 1 \times 2^1$$

$$11111111_2 = 1 \times 2^7 + 1 \times 2^6 + 1 \times 2^5 + 1 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$$

$$10010001_2 = 1 \times 2^7 + 1 \times 2^4 + 1 \times 2^0$$

4.5

$$10_2 = 1 \times 2^1 + 0 \times 2^0$$

$$1100_2 = 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 0 \times 2^0$$

$$11010010_2 = 1 \times 2^7 + 1 \times 2^6 + 1 \times 2^4 + 1 \times 2^1$$

$$13_8 = 1 \times 8^1 + 3 \times 8^0$$

$$173_8 = 1 \times 8^2 + 7 \times 8^1 + 3 \times 8^0$$

$$7171_8 = 7 \times 8^3 + 1 \times 8^2 + 7 \times 8^1 + 1 \times 8^0$$

$$1F_{16} = 1 \times 16^1 + 15 \times 16^0$$

$$F1_{16} = 15 \times 16^1 + 1 \times 16^0$$

$$FFAF_{16} = 15 \times 16^3 + 15 \times 16^2 + 10 \times 16^1 + 15 \times 16^0$$

$$10_{10} = 1010_2$$

$$\begin{array}{r} 2 \overline{) 10} - 0 \\ 2 \overline{) 5} - 1 \\ 2 \overline{) 2} - 0 \\ 1 - 1 \end{array}$$

$$256_{10} = 100000000_2$$

$$\begin{array}{r} 2 \overline{) 256} - 0 \\ 2 \overline{) 128} - 0 \\ 2 \overline{) 64} - 0 \\ 2 \overline{) 32} - 0 \\ 2 \overline{) 16} - 0 \\ 2 \overline{) 8} - 0 \\ 2 \overline{) 4} - 0 \\ 2 \overline{) 2} - 0 \\ 1 - 1 \end{array}$$

$$999_{10} = 111100111_2$$

$$\begin{array}{r} 2 \overline{) 999} - 1 \\ 2 \overline{) 499} - 1 \\ 2 \overline{) 249} - 1 \\ 2 \overline{) 124} - 0 \\ 2 \overline{) 62} - 0 \\ 2 \overline{) 31} - 1 \\ 2 \overline{) 15} - 1 \\ 2 \overline{) 7} - 1 \\ 2 \overline{) 3} - 1 \\ 1 - 1 \end{array}$$

$$\begin{array}{r}
 2 \overline{) 64} - 0 \\
 2 \overline{) 32} - 0 \\
 2 \overline{) 16} - 0 \\
 2 \overline{) 8} - 0 \\
 2 \overline{) 4} - 0 \\
 2 \overline{) 2} - 0 \\
 1 - 1
 \end{array}$$

$$\begin{array}{r}
 2 \overline{) 499} - 1 \\
 2 \overline{) 249} - 1 \\
 2 \overline{) 124} - 0 \\
 2 \overline{) 62} - 0 \\
 2 \overline{) 31} - 1 \\
 2 \overline{) 15} - 1 \\
 2 \overline{) 7} - 1 \\
 2 \overline{) 3} - 1 \\
 1 - 1
 \end{array}$$

$$0.101_2 = 1 \times 2^{-1} + 1 \times 2^{-3}$$

$$0.1101_2 = 1 \times 2^{-1} + 1 \times 2^{-2} + 1 \times 2^{-3}$$

$$0.1111_2 = 1 \times 2^{-1} + 1 \times 2^{-2} + 1 \times 2^{-3} + 1 \times 2^{-4}$$

$$0.1F_{16} = 1 \times 16^{-1} + 15 \times 16^{-2}$$

$$0.F1_{16} = 15 \times 16^{-1} + 1 \times 16^{-2}$$

$$0.ABCD = 10 \times 16^{-1} + 11 \times 16^{-2} + 12 \times 16^{-3} + 13 \times 16^{-4}$$

$$0.1_{10} = (0.0001...)_{2^{\infty}}$$

$$(0.256)_{10} = (0.010...)_{2^{\infty}}$$

$$(0.128)_{10} = (0.0010...)_{2^{\infty}}$$

$$0.1 \times 2 = 0.2 - 0$$

$$0.2 \times 2 = 0.4 - 0$$

$$0.4 \times 2 = 0.8 - 0$$

$$0.8 \times 2 = 1.6 - 1$$

$$0.6 \vdots$$

$$0.256 \times 2 = 0.512 - 0$$

$$0.512 \times 2 = 1.024 - 1$$

$$0.024 \times 2 = 0.048 - 0$$

⋮

$$0.128 \times 2 = 0.256 - 0$$

$$0.256 \times 2 = 0.512 - 0$$

$$0.512 \times 2 = 1.024 - 1$$

$$0.024 \times 2 = 0.048 - 0$$

⋮